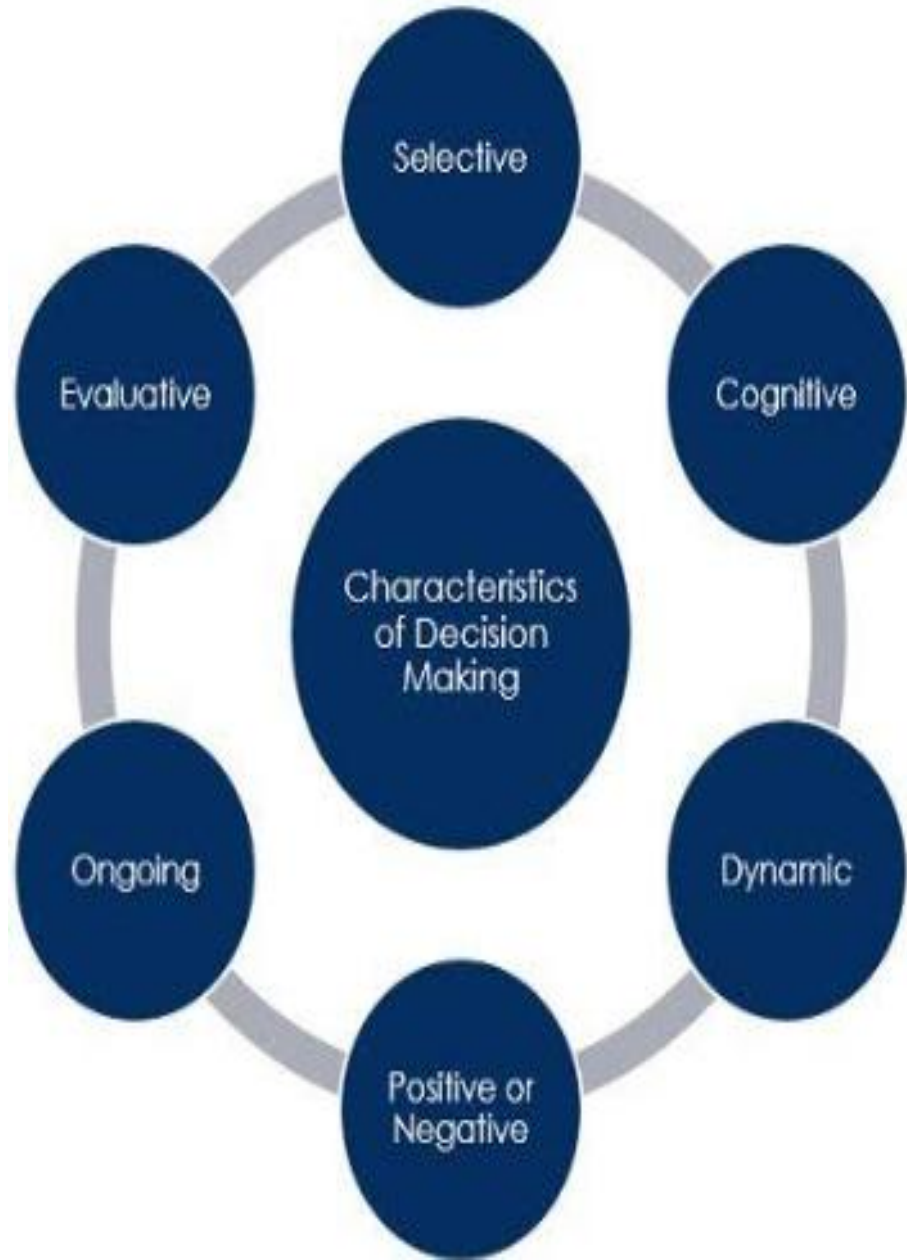




Decision Making in Management Accounting

9.1 Introduction

9.2 Key Principles in Decision Making

9.3 Decision Areas (Product Mix, Make or Buy/Outsource, Shut Down and Special Order

MBA FMA · TOPIC 9

What Is Decision Making in Management Accounting?

Definition

Decision making uses relevant costs, revenues, profitability and resource constraints to evaluate alternatives and select the best option for the organisation's objectives.

Role of Management Accounting

Management accountants are not only expected to **provide information** for decisions but also to **assist in making** those decisions spanning costing, pricing, budgeting and performance evaluation.

Four Key Decision Areas

This topic focuses on four specific decision areas within management accounting:



Product Mix

Which products to produce and in what quantities



Make or Buy

Manufacture internally or outsource to a supplier



Shut Down

Discontinue a product line, department or branch



Special Order

Accept a one-off order at a non-standard price

Relevant Revenues and Costs

Eight core principles guide relevant cost analysis in management accounting decisions.

1

Future Revenues & Costs

Only future costs matter past (sunk) costs cannot be changed by the current decision.

2

Cash Flows

Relevant items involve actual cash inflows or outflows. Non-cash items like depreciation are excluded.

3

Incremental Costs

Only the additional revenues or costs arising from choosing one alternative over another are relevant.

4

Avoidable vs Unavoidable

Avoidable costs are relevant; unavoidable costs continue regardless of the decision and are irrelevant.

Further Principles of Relevance

1

Differential Costs

The difference in total costs or revenues between alternatives helps identify the financially preferable option.

2

Opportunity Costs

The benefit sacrificed by choosing one alternative over the next best. Relevant even without a cash payment.

3

Controllable Costs

Costs a manager can influence are relevant for performance evaluation; uncontrollable costs are ignored.

4

Fixed vs Variable Costs

Variable costs are usually relevant. Fixed costs are only relevant if they **change** as a result of the decision.

Relevant vs Irrelevant Costs: A Summary

Always Relevant

- Future variable costs (materials, labour, variable overhead)
- Avoidable fixed costs
- Opportunity costs
- Incremental/differential costs
- Additional costs caused by the decision

Always Irrelevant

- Sunk costs (past costs already incurred)
- Unavoidable fixed costs
- Allocated/apportioned overheads that continue regardless
- Depreciation (non-cash)
- Committed costs

OPPORTUNITY COST

THE REAL COST OF EVERY CHOICE



Opportunity Cost is the value of the next best alternative you **give up** when you make a decision.

THE CONCEPT

Every choice has a trade-off.

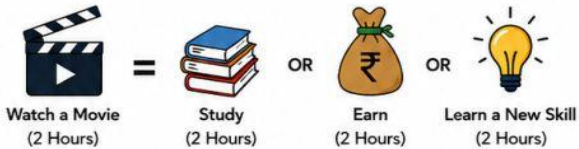
You can't have everything.

When you choose one option, you give up another.



EXAMPLE

If you spend 2 hours **watching a movie**, you give up 2 hours you could have used to **study**, **earn**, or **learn a new skill**.



WHY IT MATTERS?



Everything has a cost.
Not just in money, but in time, effort and possibilities.



Better decisions.
Helps you compare alternatives wisely.



Focus on what adds the most value.
Use your limited resources smartly.

REAL LIFE EXAMPLES



Choosing a 3-year degree
means you give up 1 year you could have spent working and earning.



Taking a job in one city
may mean giving up a better opportunity in another city.



Spending time on social media
means less time for study, exercise, reading or skill building.



Buying something you don't really need
means you could have saved or invested that money.



KEY TAKEAWAY

There's no such thing as a free choice.
Every choice you make has an opportunity cost.
Think before you choose!

“ The art of decision making is the art of choosing what you won't do.

— Economics Wisdom

The Role of Opportunity Cost

Opportunity cost is the **benefit sacrificed** by choosing one alternative over the next best. It does not involve a cash payment but is always relevant to management accounting decisions.



Make or Buy

If making a component uses capacity that could generate \$20,000 elsewhere, that \$20,000 is an opportunity cost of making.



Product Mix

Producing one product uses scarce labour hours that could have been used for a more profitable product.



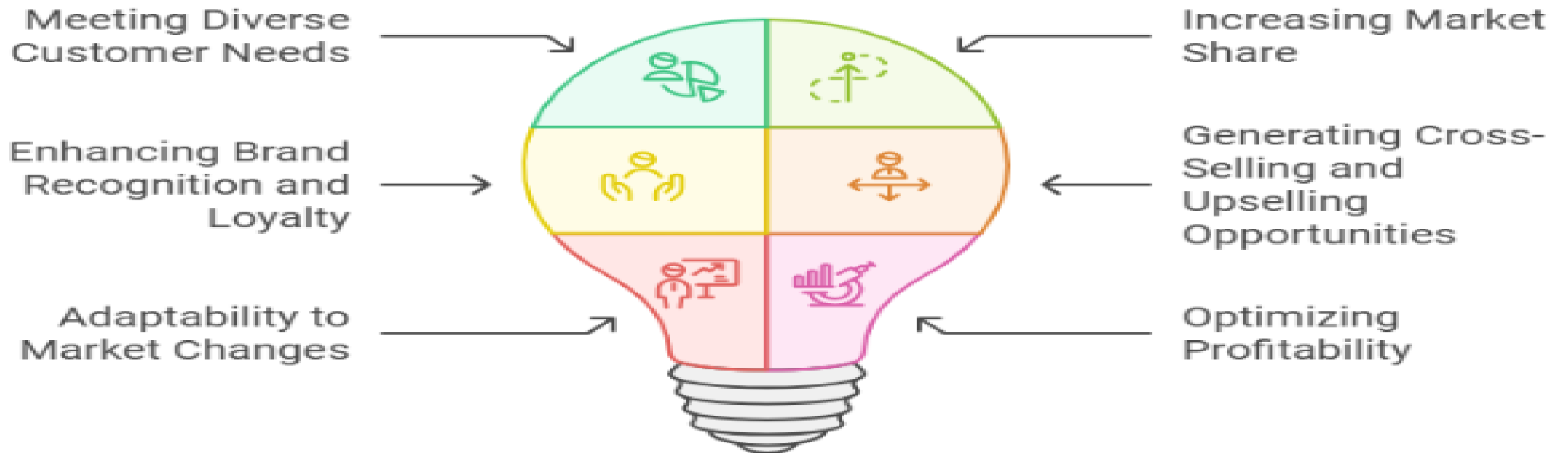
Special Order

If no spare capacity exists, accepting a special order displaces regular sales, the lost contribution is an opportunity cost.

Decision Area 1: Product Mix

When resources are limited, management must decide **which products to produce** and in what quantities to maximise total contribution from the available limiting factor.

Importance of Product Mix



Product Mix: Decision Criteria & Qualitative Factors

Decision Criteria

Select the product mix that **maximises total contribution per unit of the limiting resource**. Compare contribution per unit of limiting factor not simply contribution per unit.

Qualitative Factors

- Customer demand and preferences
- Product quality and brand image
- Strategic importance of the product
- Reliability of suppliers and production resources
- Employee skills and production capabilities
- Long-term market potential

WORKED EXAMPLE

Product Mix Example: ABC Manufacturing Ltd

ABC Manufacturing produces Products A, B and C with only **1,000 direct labour hours** available. Labour hours are the limiting factor.

Particulars	Product A	Product B	Product C
Selling price per unit	\$100	\$120	\$150
Direct materials	\$30	\$40	\$50
Direct labour	\$20	\$30	\$40
Variable production overhead	\$10	\$15	\$20
Labour hours per unit	2	1	4
Maximum monthly demand	300 units	400 units	200 units

- ❏ Allocated factory rent (\$8), administration costs (\$5) and existing depreciation (\$4) are the same for all products and are irrelevant they are unavoidable fixed costs.

WORKED EXAMPLE · STEP 1-2

Relevant Costs & Contribution per Unit

Only direct materials, direct labour and variable production overhead are relevant. Allocated fixed costs and depreciation are irrelevant.

Cost per unit	Product A	Product B	Product C
Direct materials	\$30	\$40	\$50
Direct labour	\$20	\$30	\$40
Variable production overhead	\$10	\$15	\$20
Relevant cost per unit	\$60	\$85	\$110
Selling price	\$100	\$120	\$150
Contribution per unit	\$40	\$35	\$40

Products A and C both yield \$40 contribution per unit but because labour is scarce, we must calculate **contribution per labour hour** to determine priority.

Ranking & Optimal Production Mix

Product	Contribution/unit	Labour hrs/unit	Contribution/labour hr	Rank
B	\$35	1	\$35	1st
A	\$40	2	\$20	2nd
C	\$40	4	\$10	3rd

Priority: **Product B** → **Product A** → **Product C**

Product	Units	Hrs/unit	Hours used	Cumulative hrs
B	400	1	400	400
A	300	2	600	1,000
C	0	4	0	1,000

Total Contribution & Why Fixed Costs Are Excluded

Total Contribution

Product	Units	Contribution/unit	Total
B	400	\$35	\$14,000
A	300	\$40	\$12,000
C	0	\$40	\$0
Total	700		\$26,000

\$26,000

Total Contribution

Optimal mix: 400 units of B + 300 units of A

Why Exclude Fixed Costs?

Allocated factory rent, administration costs and existing depreciation are **unavoidable fixed costs**. They will be incurred regardless of which products are made. Including them would distort the decision.

1,000

Labour Hours Used

All available hours fully utilised



Decision Area 2: Make or Buy / Outsource

Should the organisation manufacture a component internally or purchase it from an external supplier? The answer depends on relevant avoidable costs and opportunity costs not total accounting costs.

Make or Buy: Decision Criteria & Qualitative Factors

Decision Criteria

Compare the **relevant avoidable cost of making** with the **purchase price**. Include any opportunity cost of capacity used internally. Buy if the relevant cost of making exceeds the relevant cost of buying.

Qualitative Factors

- Quality and reliability of the supplier
- Supplier delivery reliability
- Confidentiality and intellectual property
- Long-term supplier dependence
- Impact on employees and internal capabilities
- Strategic control over the activity

WORKED EXAMPLE

Make or Buy Example: XYZ Manufacturing Ltd

XYZ currently makes 10,000 units of Component X per year. An external supplier offers to supply at **\$28 per unit**.

Cost per unit	\$	Treatment
Direct materials	10.00	Relevant — avoided if bought
Direct labour	8.00	Relevant — avoided if bought
Variable production overhead	4.00	Relevant — avoided if bought
Fixed production overhead (40% avoidable)	2.80	Partly relevant
Fixed production overhead (60% unavoidable)	4.20	Irrelevant — continues either way
Allocated administration costs	3.00	Irrelevant — continues either way
Relevant cost of making per unit	24.80	

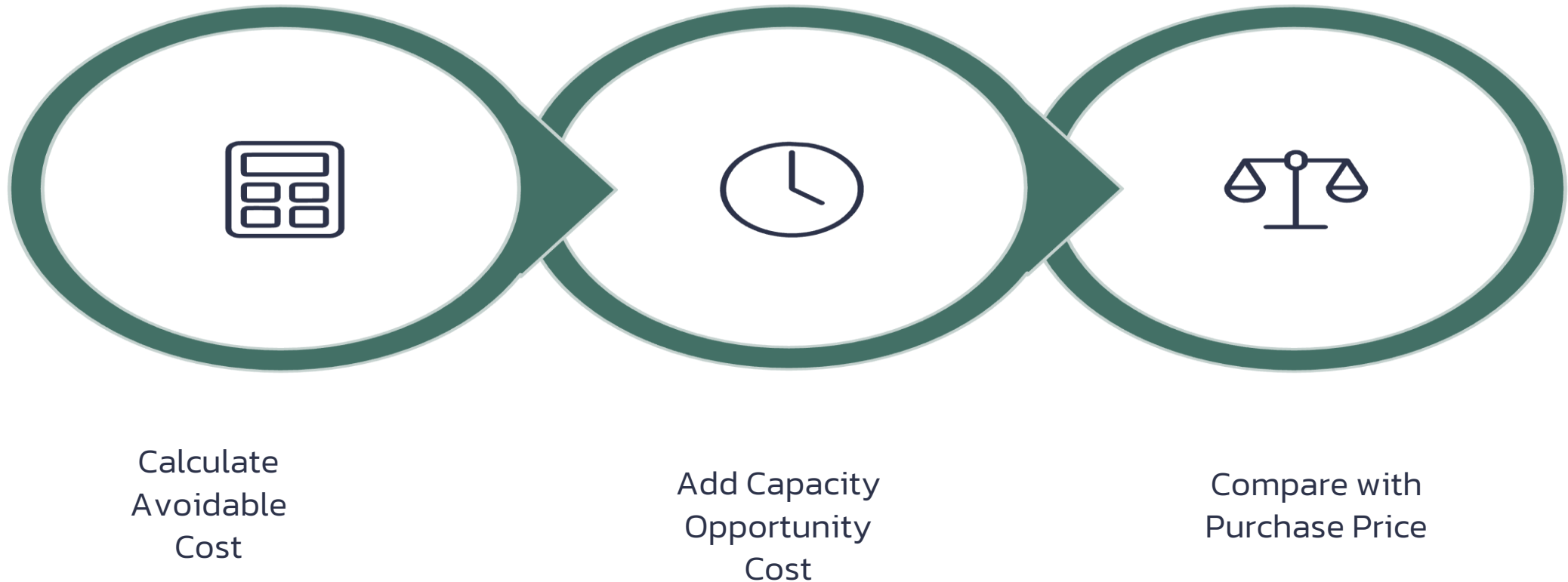
Make or Buy: Final Decision with Opportunity Cost

	Make	Buy
Relevant production cost (10,000 units)	\$248,000	\$280,000
Opportunity cost of capacity	\$20,000	—
Relevant economic cost	\$268,000	\$280,000

Buying releases capacity generating an additional contribution of **\$20,000**, an opportunity cost of making. Even so, making (\$268,000) remains cheaper than buying (\$280,000).

- ✔ XYZ Manufacturing Ltd should continue to **MAKE** the component a financial advantage of **\$12,000** over buying.

Make or Buy: The Decision Framework

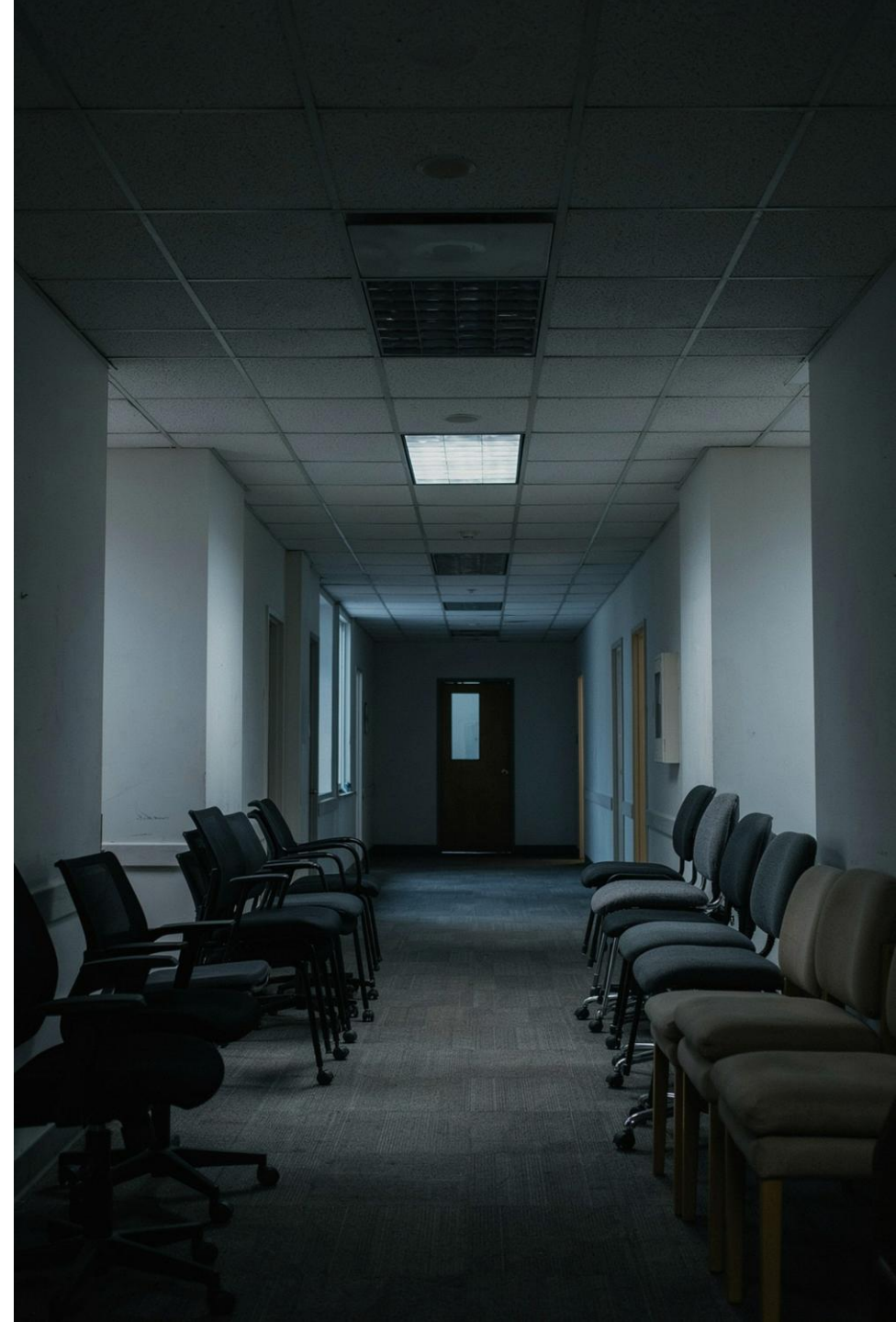


The decision is not based on total accounting cost but on the **relevant economic cost** of each alternative, including any opportunity cost of the capacity released or consumed.

Decision Area 3: Shut Down

Should a product line, department, branch or business be **discontinued**?

An accounting loss does not automatically justify shutdown
contribution and avoidable costs must be assessed.



Shutdown: Decision Criteria & Qualitative Factors

Decision Criteria

Shut down only if **avoidable costs saved exceed the contribution lost** from discontinuing, after considering any opportunity costs. An accounting loss alone is insufficient justification.

Qualitative Factors

- Impact on customers and reputation
- Effect on other products or operations
- Employee and community consequences
- Future growth potential
- Strategic importance of the operation
- Possibility of improving efficiency or profitability

WORKED EXAMPLE

Shutdown Example: ABC Manufacturing Ltd

ABC produces four product lines. Product D reports an accounting loss of \$7,000. Should any line be discontinued?

Particulars	Product A	Product B	Product C	Product D
Sales revenue	\$240,000	\$180,000	\$150,000	\$120,000
Variable costs	(130,000)	(100,000)	(90,000)	(77,000)
Contribution	\$110,000	\$80,000	\$60,000	\$43,000
Specific fixed costs	(25,000)	(20,000)	(35,000)	(30,000)
Allocated general fixed costs	(30,000)	(25,000)	(20,000)	(20,000)
Reported profit/(loss)	\$55,000	\$35,000	\$5,000	(\$7,000)



Allocated general fixed costs are **irrelevant** they continue regardless of any shutdown decision.

Shutdown Decision: Net Financial Effect

Key formula: **Net effect of shutdown** = **Avoidable specific fixed costs** – **Contribution lost** ± **Opportunity benefit**

Product	Contribution	Avoidable Fixed Costs	Other Benefit	Net Effect	Decision
A	\$110,000	\$25,000	—	(\$85,000)	Continue
B	\$80,000	\$20,000	—	(\$60,000)	Continue
C	\$60,000	\$35,000	+\$18,000	(\$7,000)	Continue
D	\$43,000	\$30,000	—	(\$13,000)	Continue

- ✔ None of the four product lines should be shut down. Although Product D reports an accounting loss, it generates a positive contribution of \$43,000. Shutting it down would reduce company profit by \$13,000.

Why Product D Should Not Be Shut Down

The Accounting Loss Is Misleading

Product D's reported loss of \$7,000 includes \$20,000 of **allocated general fixed costs** that will continue regardless. These are irrelevant to the shutdown decision.

The Relevant Analysis

Product D generates a **contribution of \$43,000** and avoids \$30,000 of specific fixed costs if continued. Shutting it down would reduce total company profit by **\$13,000** the opposite of the intended outcome.



Never base a shutdown decision solely on reported accounting profit or loss. Always assess contribution and avoidable costs.

Decision Area 4: Special Orders

Should the organisation accept a **one-off order at a non-standard price?** The key is whether incremental revenue exceeds incremental costs not whether the price is below the normal selling price.



Special Order: Decision Criteria & Qualitative Factors

Decision Criteria

Accept if **incremental revenue exceeds incremental/avoidable costs**. If spare capacity is unavailable, include the opportunity cost of displaced regular sales.

Qualitative Factors

- Effect on existing customers and normal prices
- Customer relationship and future business potential
- Product quality and brand reputation
- Ability to meet delivery requirements
- Risk of the special price becoming a benchmark
- Strategic importance of the customer/order

WORKED EXAMPLE

Special Order Example: ABC Manufacturing Ltd

ABC sells Product X at **\$80 per unit** and operates at **80% capacity**. A foreign customer offers to buy 2,000 units at **\$55 per unit**. The order will not affect existing customers or normal pricing.

Cost per unit	\$	Treatment
Direct materials	18	Relevant
Direct labour	12	Relevant
Variable production overhead	8	Relevant
Fixed production overhead	15	Irrelevant — incurred regardless
Allocated administration costs	5	Irrelevant — incurred regardless
Additional packaging	2	Relevant — caused by the order
Relevant cost per unit	40	

Special Order: Incremental Profit Calculation

	Per unit	2,000 units
Special order revenue	\$55	\$110,000
Direct materials	(\$18)	(\$36,000)
Direct labour	(\$12)	(\$24,000)
Variable production overhead	(\$8)	(\$16,000)
Additional packaging	(\$2)	(\$4,000)
Incremental profit	\$15	\$30,000

- ✓ ABC Manufacturing Ltd should **ACCEPT** the special order. It generates an incremental profit of \$15 per unit **\$30,000 in total**. The price of \$55 being below the normal price of \$80 is irrelevant; the relevant comparison is price vs. incremental cost.

The Golden Rule of Special Orders

Price vs. Incremental Cost — not Price vs. Normal Price

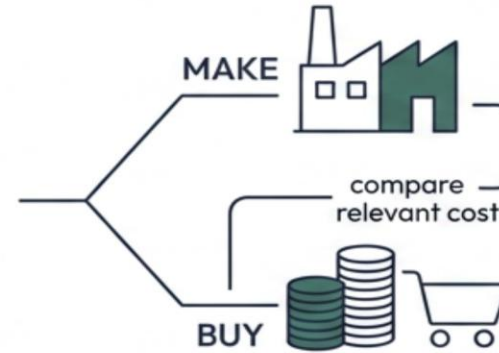
The fact that the special order price of **\$55** is below the normal selling price of **\$80** does not automatically make the order unacceptable. What matters is whether the price covers the **relevant incremental costs** and here it does, by \$15 per unit.

Comparing the Four Decision Areas



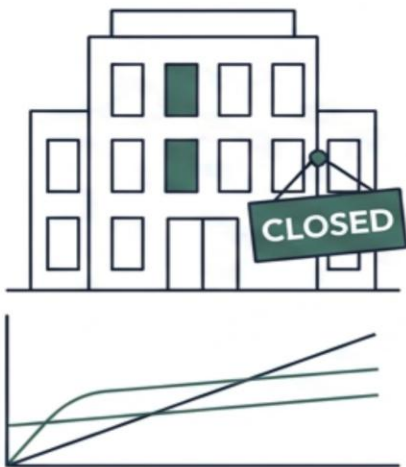
PRODUCT MIX

Maximise contribution per unit of limiting factor



MAKE OR BUY

Compare relevant cost of making vs purchase price plus opportunity cost



SHUT DOWN

Shut down only if avoidable costs saved exceed contribution lost

SPECIAL ORDER

Accept if incremental revenue exceeds incremental costs



Shutdown Decision: The Decision Framework

Calculate Contribution

Compute product line contribution margin.

Compare & Decide

Shut down if savings exceed lost contribution.



Identify Avoidable Costs

List avoidable fixed costs and opportunity benefits.

A product line reporting an accounting loss should only be discontinued if the **costs saved exceed the contribution lost**. Allocated fixed costs must be excluded from this analysis.

Qualitative Factors Matter Too

Financial analysis provides the foundation, but qualitative factors can override a purely quantitative recommendation.

Reputation & Brand

Shutting down a product line or accepting a low-price special order may damage brand perception or signal weakness to competitors.

Supplier Dependence

Outsourcing a component may create long-term dependence on a single supplier, increasing vulnerability to price rises or supply disruptions.

Employee & Community Impact

Discontinuing operations affects staff morale, employment and community relationships factors that carry strategic and ethical weight.

Strategic Importance

A product or operation may be loss-making today but strategically vital for future growth, market position or customer retention.

Key Takeaways

1 Only future, incremental, avoidable costs are relevant

Sunk costs, allocated overheads and unavoidable fixed costs must be excluded from all decision analyses.

3 Opportunity cost is always relevant

The benefit foregone by choosing one alternative must be included in the relevant cost comparison.

2 Contribution per limiting factor drives product mix

Rank products by contribution per unit of scarce resource not by contribution per unit alone.

4 Accounting loss $\neq$ shut down; price below normal $\neq$ reject

Decisions must be based on relevant economic analysis, not reported accounting figures.

A photograph of a wooden desk with an open notebook, a pen, and a cup of coffee. The notebook is open to a blank page, and the pen is resting on it. A white cup of coffee sits on a saucer next to the notebook. The scene is lit with warm, natural light.

Topic 9 Summary

Decision making in management accounting requires identifying what is truly relevant, applying the correct decision framework, and balancing financial rigour with strategic and qualitative judgement.

Product Mix

Maximise contribution per limiting factor

Make or Buy

Compare relevant economic costs including opportunity cost

Shut Down

Shut down only if savings exceed contribution lost

Special Order

Accept if incremental revenue exceeds incremental cost